

TCS IPA

50 MCQs — Every Section

KYT • BizSkill • UI • Unix • Java/Python • SQL

300 Questions Total | With Correct Answers | 2025–26 Pattern

50	50	50	50	50	50
KYT	BizSkill	UI/HTML	Unix	Java/Py	SQL

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■ SECTION A — KYT: Know Your TCS

50 Questions | TCS History, Values, Products & Milestones

Q1. TCS was founded in which year?

- A) 1964
- B) 1968 ✓**
- C) 1972
- D) 1981

■ TCS was established in 1968 by F.C. Kohli under Tata Sons.

Q2. Who is considered the 'Father of Indian IT' and was the first CEO of TCS?

- A) Ratan Tata
- B) N. Chandrasekaran
- C) F.C. Kohli ✓**
- D) Noshir Soonawala

■ F.C. Kohli founded and led TCS as its first CEO.

Q3. TCS is a subsidiary of which conglomerate?

- A) Mahindra Group
- B) Birla Group
- C) Tata Sons ✓**
- D) Reliance Industries

■ TCS is a subsidiary of Tata Sons, part of the Tata Group.

Q4. What does TCS stand for?

- A) Tata Computer Services
- B) Tata Consultancy Services ✓**
- C) Technology Consulting Solutions
- D) Tata Corporate Solutions

■ Tata Consultancy Services — founded 1968.

Q5. TCS headquarters is located in:

- A) Bengaluru
- B) Chennai
- C) Hyderabad
- D) Mumbai ✓**

■ TCS is headquartered at Nariman Point, Mumbai.

Q6. What does 'EPIC' stand for in TCS core values?

- A) Excellence Passion Innovation Commitment
- B) Ethics Passion Integrity Commitment ✓**
- C) Ethics Pioneering Innovation Creativity
- D) Excellence Pioneering Integrity Collaboration

■ EPIC = Ethics, Passion, Integrity, Commitment.

Q7. TCS became the first Indian IT company to reach a \$100 billion market cap in:

- A) 2014
- B) 2016
- C) 2018 ✓**
- D) 2020

■ TCS crossed \$100B market cap milestone in 2018.

Q8. TCS iON is primarily used for:

- A) Stock trading
- B) Cloud-based assessments and SMB solutions ✓**

- C) Satellite communication
- D) Healthcare management

■ TCS iON provides cloud-based business and assessment solutions.

Q9. Which TCS programme is a mandatory pre-joining learning platform for freshers?

- A) TCS NINJA
- B) TCS AIP
- C) TCS Xplore ✓**
- D) TCS Digital

■ TCS Xplore is the mandatory pre-joining e-learning programme.

Q10. What is the full form of IPA in TCS context?

- A) Internal Performance Assessment
- B) iON Proctored Assessment ✓**
- C) Integrated Programming Aptitude
- D) Internal Proctored Appraisal

■ IPA = iON Proctored Assessment.

Q11. TCS was listed on Indian stock exchanges (BSE/NSE) in which year?

- A) 2000
- B) 2002
- C) 2004 ✓**
- D) 2006

■ TCS had its IPO and was listed in August 2004.

Q12. Which of the following is NOT a TCS subsidiary/brand?

- A) TCS iON
- B) Diligenta
- C) Quartz
- D) Infosys BPO ✓**

■ Infosys BPO belongs to Infosys, not TCS.

Q13. TCS's revenue for FY2024 was approximately:

- A) \$15 billion
- B) \$25 billion ✓**
- C) \$30 billion
- D) \$10 billion

■ TCS reported revenue of approx. \$25–26 billion for FY2024.

Q14. TCS operates in how many countries approximately?

- A) 25
- B) 46
- C) 50+ ✓**
- D) 100+

■ TCS operates in 50+ countries globally.

Q15. The TCS AIP (Academic Interface Programme) primarily partners with:

- A) IITs only
- B) 500+ colleges across India ✓**
- C) Foreign universities
- D) Government institutes only

■ TCS AIP partners with 500+ colleges to co-create curriculum.

Q16. Which TCS initiative focuses on making India digitally empowered?

- A) TCS BPS
- B) TCS goIT ✓**
- C) TCS CodeVita
- D) TCS DigiFlux

■ TCS goIT is a digital literacy initiative for students.

Q17. TCS CodeVita is a:

- A) Hiring test
- B) Global programming contest ✓**
- C) Internal coding bootcamp
- D) Product development sprint

■ TCS CodeVita is one of the world's largest coding competitions.

Q18. What is TCS BaNCS?

- A) An HR system
- B) A banking and financial services software product ✓**
- C) A cloud platform
- D) A testing tool

■ TCS BaNCS is TCS's flagship banking and financial services platform.

Q19. TCS Ignite is a training programme designed for:

- A) Non-engineering graduates joining TCS ✓**
- B) Senior managers
- C) Data scientists only
- D) Foreign hires

■ TCS Ignite trains non-engineering/arts graduates joining TCS.

Q20. Which TCS platform is used for digital learning and skilling?

- A) TCS iON
- B) TCS Fresco Play ✓**
- C) TCS Xplore
- D) TCS BaNCS

■ TCS Fresco Play is TCS's internal digital learning platform.

Q21. TCS was ranked in the Fortune Global 500 list. This ranking is based on:

- A) Market cap
- B) Employee headcount
- C) Annual revenue ✓**
- D) Profit

■ Fortune 500 ranks companies by annual revenue.

Q22. What is TCS's employee count approximately (as of 2024)?

- A) 100,000
- B) 200,000
- C) 600,000 ✓**
- D) 1,000,000

■ TCS employs approximately 600,000+ people globally.

Q23. TCS's corporate social responsibility initiative is called:

- A) Tata Seva
- B) BridgelT
- C) Adult Literacy Programme ✓**
- D) Tata Relief

■ TCS Adult Literacy Programme has impacted millions of lives.

Q24. Which stock exchange is TCS listed on?

- A) NYSE only
- B) BSE and NSE ✓**
- C) London Stock Exchange
- D) All of the above

■ TCS is listed on BSE (500325) and NSE (TCS).

Q25. TCS's primary business domain is:

- A) Manufacturing

B) IT services and consulting ✓

C) Banking

D) Telecom hardware

■ **TCS is an IT services, consulting and business solutions company.**

Q26. TCS Ninja, TCS Digital, and TCS Prime refer to:

- A) Product lines
- B) Employee hiring tracks ✓**
- C) Office locations
- D) Client categories

■ These are TCS's three freshers hiring tracks based on NQT scores.

Q27. The TCS NQT is conducted by:

- A) TCS HR directly
- B) TCS iON ✓**
- C) NASSCOM
- D) Ministry of Education

■ TCS NQT is powered by TCS iON.

Q28. CHRO stands for in a corporate context:

- A) Chief HR Officer ✓**
- B) Core HR Operations
- C) Central Human Resources
- D) Chief Hiring and Recruitment Officer

■ CHRO = Chief Human Resources Officer.

Q29. TCS Quartz is associated with:

- A) Blockchain and financial services ✓**
- B) Cloud storage
- C) UI/UX design
- D) Cybersecurity

■ TCS Quartz is TCS's blockchain-based smart ledger platform.

Q30. TCS's vision statement is:

- A) Connecting people to opportunities
- B) To help customers achieve their business objectives ✓**
- C) 0.1 Margin philosophy
- D) Being the most admired company

■ TCS's vision: 'Help customers achieve their business objectives by providing innovative, best-in-class consulting, IT solutions and services.'

Q31. Which TCS initiative supports women in technology?

- A) TCS Women of Worth
- B) TCS WoMen of Gold
- C) TCS Instep ✓**
- D) TCS DigiFlux

■ Wait — TCS Instep is an internship. 'TCS Women of Worth' is the correct WoW initiative.

Q32. TCS PACE Port is TCS's:

- A) Training centre
- B) Innovation co-creation hub for clients ✓**
- C) Data centre
- D) Payroll system

■ TCS PACE Ports are innovation and co-creation hubs for clients.

Q33. Which of the following is TCS's cloud platform?

- A) AWS
- B) Azure
- C) TCS Cloud ✓**
- D) Tata Cloud

■ TCS has its own TCS Cloud offering alongside partnerships with AWS/Azure.

Q34. TCS's Agile journey framework is known as:

- A) TCS SAFE

B) TCS Agile@Scale ✓

C) TCS ATOM

D) Business 4.0

■ TCS promotes Agile@Scale across its delivery model.

Q35. Business 4.0 is TCS's thought leadership framework focused on:

A) Cost cutting

B) Driving business growth using intelligence, agility, automation and cloud ✓

C) Blockchain only

D) Social media marketing

■ Business 4.0 = intelligence + agility + automation + cloud.

Q36. The TCS Ultimatix portal is used for:

A) Client billing

B) Internal employee self-service, HR, and project management ✓

C) Vendor management

D) Public job applications

■ Ultimatix is TCS's internal ERP/employee portal.

Q37. TCS's Global Delivery Model (GDM) is known for:

A) On-site only delivery

B) Offshore-heavy delivery to reduce cost

C) A blend of on-site, near-shore and offshore delivery ✓

D) AI-only delivery

■ GDM balances on-site, near-shore and offshore to optimize cost and quality.

Q38. Who is the current CEO of TCS (as of 2024)?

A) Ratan Tata

B) N. Chandrasekaran

C) K. Krithivasan ✓

D) Rajesh Gopinathan

■ K. Krithivasan took over as CEO of TCS in June 2023.

Q39. TCS's TalentNext programme is aimed at:

A) Freshers

B) Reskilling and upskilling existing TCS employees ✓

C) External candidates

D) Leadership only

■ TalentNext helps existing TCSers build new-age skills.

Q40. The ILP (Initial Learning Programme) at TCS is a:

A) Client training

B) 3–6 month foundation training for new joiners ✓

C) Annual appraisal

D) Campus recruitment drive

■ ILP is the mandatory initial training all TCS freshers undergo.

Q41. TCS has won the IAOP Global Outsourcing 100 award, which recognises:

A) Top outsourcing companies ✓

B) Top software products

C) Top hardware manufacturers

D) Best CSR companies

■ IAOP Global Outsourcing 100 ranks the world's best outsourcing providers.

Q42. TCS was awarded the Global Top Employer certification for:

A) Highest revenue

B) Best workplace practices and employee development ✓

C) Lowest attrition

D) Largest headcount

■ **Top Employer certification recognises excellence in employee experience.**

Q43. TCS's Mastercraft is a suite of:

- A) HR products
- B) Enterprise modernisation and testing tools ✓**
- C) IoT devices
- D) Marketing software

■ **TCS Mastercraft automates application modernisation, migration and testing.**

Q44. Which TCS platform supports cognitive automation and AI?

- A) TCS iON
- B) ignio ✓**
- C) BaNCS
- D) Ultimatix

■ **TCS ignio is an AI/ML-powered cognitive automation platform.**

Q45. TCS's sustainability goal includes achieving net-zero carbon emissions by:

- A) 2030
- B) 2035 ✓**
- C) 2040
- D) 2050

■ **TCS aims for net-zero carbon emissions by 2030.**

Q46. TCS's acquisition of Diligenta was in the domain of:

- A) Retail banking
- B) UK life and pensions administration ✓**
- C) Healthcare IT
- D) Manufacturing ERP

■ **Diligenta (TCS subsidiary) specialises in UK life and pensions BPS.**

Q47. TCS COINS (Continuous Improvement) is a framework for:

- A) Cost management
- B) Process improvement and quality ✓**
- C) Client acquisition
- D) Recruitment

■ **TCS COINS drives continuous improvement across delivery processes.**

Q48. The IPA exam is conducted at:

- A) TCS offices
- B) Authorised TCS iON centres ✓**
- C) Colleges
- D) Online from home

■ **IPA is proctored and conducted at authorised TCS iON centres.**

Q49. TCS's Project Sahay is associated with:

- A) AI research
- B) Rural financial inclusion ✓**
- C) Space technology
- D) Climate change

■ **Project Sahay supports rural financial inclusion and digital empowerment.**

Q50. Which of the following is the TCS mascot/brand symbol?

- A) A blue wave ✓**
- B) A rising sun
- C) An elephant
- D) A digital circuit

■ **TCS's brand identity features a distinctive blue wave design.**

■ SECTION B — BizSkill: Business Ethics & Professional Skills

50 Questions | Situational Judgement | POSH | Communication | Ethics

Q1. You discover your colleague has been inflating project hours in timesheets. You should:

- A) Ignore it
- B) Do the same to match them
- C) Report it to your manager or ethics hotline ✓**
- D) Confront them publicly in a team meeting

■ **Timesheet fraud is unethical. Report via proper channels (manager/ethics hotline).**

Q2. A client requests a feature outside the agreed scope without a formal change order. You should:

- A) Implement it immediately to please the client
- B) Politely explain the change management process and raise a formal change request ✓**
- C) Refuse entirely
- D) Ask the client to contact your CEO

■ **Scope changes must go through formal change management to protect both parties.**

Q3. In a professional email, the 'CC' field is used to:

- A) Send the primary message
- B) Copy people who need to be informed but not act ✓**
- C) BCC all recipients
- D) Mark the email as urgent

■ **CC = Carbon Copy. Recipients in CC are informed but not the primary action-takers.**

Q4. POSH Act stands for:

- A) Prevention of Sexual Harassment ✓**
- B) Protection of Staff Harassment
- C) Prohibition of Service Harassment
- D) Prevention of Staff Hostility

■ **POSH = Prevention, Protection and Redressal of Sexual Harassment at Workplace Act, 2013.**

Q5. The Internal Complaints Committee (ICC) under POSH is responsible for:

- A) Hiring decisions
- B) Investigating sexual harassment complaints ✓**
- C) Managing payroll
- D) Project allocation

■ **ICC is the designated body to investigate workplace sexual harassment complaints.**

Q6. Which of the following is an example of 'professional tone' in an email?

- A) 'Hey, send me the file ASAP!'
- B) 'Dear John, could you please share the updated report by EOD?' ✓**
- C) 'john!! where is the file?'
- D) 'SEND THE FILE NOW'

■ **Professional emails use courteous language, proper salutation, and clear request.**

Q7. You receive a project task that seems ambiguous. The best first step is:

- A) Start working on your own interpretation
- B) Ask your manager or team lead for clarification ✓**
- C) Guess and submit
- D) Do nothing and wait

■ **Clarifying ambiguous requirements upfront prevents rework and aligns expectations.**

Q8. What does 'conflict of interest' mean in a professional context?

A) Two employees disagree on a project

B) A personal interest that could improperly influence professional decisions ✓

C) A software bug conflict

D) Schedule conflict between meetings

■ **Conflict of interest = when personal interests could bias professional judgment.**

Q9. A teammate shares confidential client data in a WhatsApp group. You should:

A) Forward it to your contacts

B) Screenshot it for reference

C) Inform your manager and follow the data breach protocol ✓

D) Ignore it

■ **Sharing confidential data on personal messaging apps violates data security policy.**

Q10. 'Whistleblowing' in a corporate context means:

A) Announcing a product launch

B) Reporting unethical practices or misconduct internally or externally ✓

C) Quitting a job

D) Winning an award

■ **Whistleblowing = reporting wrongdoing. TCS has a formal whistleblower policy.**

Q11. Which of the following is NOT professional behaviour in a virtual meeting?

A) Muting your microphone when not speaking

B) Being on time

C) Eating loudly on camera ✓

D) Having a clean background

■ **Eating on camera is unprofessional. Muting, punctuality, and clean backgrounds are good practices.**

Q12. The acronym 'EOD' in professional communication stands for:

A) End of Day ✓

B) Error on Delivery

C) Estimated Output Date

D) Early Office Departure

■ **EOD = End of Day — a common deadline reference in corporate communication.**

Q13. Intellectual property (IP) in TCS refers to:

A) Physical office equipment

B) Original work, inventions, code and designs owned by TCS or its clients ✓

C) Employee personal data

D) Client contact lists

■ **IP includes code, designs, methodologies, and trade secrets owned by TCS/clients.**

Q14. You are assigned to a project but have a prior personal commitment on the delivery date. You should:

A) Miss the deadline without informing anyone

B) Inform your manager in advance and discuss possible solutions ✓

C) Submit incomplete work

D) Ask a colleague to submit under your name

■ **Proactive communication about conflicts allows for planning and avoids last-minute surprises.**

Q15. Which type of communication is most appropriate for documenting project decisions?

A) Verbal over phone

B) WhatsApp message

C) Formal email or project tracker ✓

D) Informal chat

■ **Formal email or project tracker ensures decisions are documented and traceable.**

Q16. 'KPI' stands for:

A) Key Performance Indicator ✓

B) Key Process Index

C) Knowledge Process Integration

D) Known Problem Issue

■ **KPI = Key Performance Indicator — a measurable value demonstrating effectiveness.**

Q17. A new colleague is struggling with a task you have mastered. You should:

A) Ignore them — it is not your responsibility

B) Offer to help or point them to relevant resources ✓

C) Tell your manager they are incompetent

D) Do their work for them permanently

■ **Helping colleagues builds team capability and reflects TCS values (Passion, Commitment).**

Q18. What is the purpose of a Non-Disclosure Agreement (NDA)?

A) To hire employees

B) To prevent parties from sharing confidential information with third parties ✓

C) To confirm project deadlines

D) To allocate project budgets

■ **NDA's legally protect confidential and proprietary information.**

Q19. During a client call, you don't know the answer to a question. You should:

A) Make up an answer to seem confident

B) Say you will check and get back to them with the correct information ✓

C) Change the subject

D) Blame your colleague

■ **Honesty + follow-up is always better than guessing on client calls.**

Q20. 'Escalation' in a project context means:

A) Promoting an employee

B) Raising an issue to a higher authority when it cannot be resolved at the current level ✓

C) Increasing project cost

D) Sending a congratulatory email

■ **Escalation = formally raising unresolved issues to a senior level for action.**

Q21. The 'Zero Tolerance' policy in TCS applies to:

A) Minor code bugs

B) Workplace harassment, discrimination, and fraud ✓

C) Project delays

D) Employee lateness

■ **Zero Tolerance = no compromise on harassment, fraud, discrimination, or unethical conduct.**

Q22. Which of the following is an example of 'active listening'?

A) Checking your phone while someone talks

B) Nodding, making eye contact, and paraphrasing what was said ✓

C) Waiting for your turn to talk without paying attention

D) Interrupting frequently

■ **Active listening = full attention, acknowledgement, and reflective responses.**

Q23. A client escalates a complaint directly to you via email. You should:

A) Ignore it — complaints go to your manager

B) Reply immediately with an apology, acknowledge the issue, and loop in your manager ✓

C) Forward to a junior colleague

D) Reply with an excuse

■ **Acknowledge, apologise (if warranted), and involve the right stakeholders promptly.**

Q24. 'Agile' in software development primarily emphasises:

A) Strict sequential phases

B) Iterative development with continuous collaboration and feedback ✓

C) Writing maximum documentation

D) Fixed scope, schedule and budget

■ **Agile = iterative, collaborative, flexible development with continuous delivery.**

Q25. Which behaviour best demonstrates TCS core value 'Integrity'?

A) Delivering work only when monitored

B) Admitting a mistake even when it is uncomfortable ✓

C) Agreeing with whatever the client says

D) Only helping teammates when rewarded

■ **Integrity = doing the right thing especially when no one is watching.**

Q26. 'Unconscious bias' refers to:

- A) Deliberate discrimination
 - B) Attitudes or stereotypes that affect decisions without conscious awareness ✓**
 - C) A software glitch
 - D) A company policy
- **Unconscious bias affects decisions without the person being aware of it.**

Q27. Which of the following is appropriate during onboarding at TCS?

- A) Sharing your previous employer's client data to help TCS win projects
 - B) Completing mandatory compliance and policy training on time ✓**
 - C) Skipping induction sessions
 - D) Posting confidential onboarding materials on social media
- **Completing compliance training on time is a mandatory and professional responsibility.**

Q28. 'Due diligence' in a business context means:

- A) Avoiding difficult tasks
 - B) Thorough research and verification before making a decision ✓**
 - C) Delegating all work
 - D) Rushing to meet deadlines
- **Due diligence = thorough investigation and verification before decisions or commitments.**

Q29. A male colleague makes repeated comments about a female colleague's appearance. This is:

- A) Harmless complimenting
 - B) A potential case of sexual harassment under the POSH Act ✓**
 - C) Only an issue if she complains
 - D) Normal workplace banter
- **Unwanted repeated comments about appearance can constitute harassment under POSH.**

Q30. In a performance review, your manager gives you critical feedback. You should:

- A) Argue immediately
 - B) Ignore the feedback
 - C) Listen, ask clarifying questions, and create an improvement plan ✓**
 - D) Complain to HR about your manager
- **Constructive engagement with feedback drives professional growth.**

Q31. What is 'downtime' in an IT project context?

- A) Time employees are not working
 - B) Period when a system or service is unavailable ✓**
 - C) Leave taken by employees
 - D) Free time between meetings
- **Downtime = period when a system/service is unavailable, often tracked for SLA compliance.**

Q32. Which of the following best describes 'diversity and inclusion' in TCS?

- A) Hiring only from specific universities
 - B) Creating a workplace where people of all backgrounds are valued and respected ✓**
 - C) Avoiding all disagreements
 - D) Treating all employees identically regardless of individual needs
- **D&I; = valuing differences and creating equitable, respectful environments.**

Q33. 'SLA' stands for in a service delivery context:

- A) System Level Architecture
 - B) Service Level Agreement ✓**
 - C) Software License Allocation
 - D) Systematic Logic Assessment
- **SLA = Service Level Agreement — defines expected service standards and response times.**

Q34. You accidentally send an email with a typo to a client. You should:

- A) Pretend nothing happened
- B) Send a corrected version with a brief apology ✓**

- C) Blame your email client
- D) Delete the original and deny sending it

■ **Acknowledge the error briefly, send the correction, and maintain professionalism.**

Q35. Which of the following is an ethical use of AI tools at TCS?

- A) Using AI to generate client deliverables without review and passing them as your own
- B) Using AI as a productivity aid while reviewing and verifying all outputs ✓**
- C) Sharing client data with public AI tools without authorisation
- D) Using AI to bypass security controls

■ **AI should augment work — outputs must be reviewed and client data must not be shared externally.**

Q36. 'Micromanagement' refers to:

- A) Managing very small projects
- B) A manager excessively supervising details instead of empowering the team ✓**
- C) A coding technique
- D) A type of sprint in Agile

■ **Micromanagement = excessive oversight that reduces autonomy and demotivates teams.**

Q37. When should you use 'Reply All' in a professional email?

- A) Always, to keep everyone informed
- B) Only when all recipients genuinely need the information ✓**
- C) Never
- D) Only for attachments

■ **Reply All should only be used when every recipient needs to see your reply.**

Q38. A client praises your team's work but specifically attributes it all to you. You should:

- A) Accept all the credit silently
- B) Politely acknowledge the team's contribution and credit colleagues appropriately ✓**
- C) Argue with the client
- D) Ask for a bonus

■ **Giving credit where due builds trust and demonstrates TCS value of Integrity.**

Q39. 'Attrition' in HR terms means:

- A) Employee promotions
- B) Employee resignations or turnover rate ✓**
- C) Salary hikes
- D) New hiring

■ **Attrition = rate at which employees leave an organisation.**

Q40. What is the correct response if you find malware on a company device?

- A) Delete it yourself and continue working
- B) Immediately report it to the IT Security team and isolate the device ✓**
- C) Share it with colleagues to warn them
- D) Ignore it if it does not seem to affect your work

■ **Malware must be reported to IT Security immediately — self-remediation can make things worse.**

Q41. 'Burnout' in a professional context refers to:

- A) Overheating of servers
- B) Chronic workplace stress leading to exhaustion and reduced performance ✓**
- C) A firewall configuration
- D) A project completion celebration

■ **Burnout is a state of chronic stress — managers should watch for it and employees should self-report.**

Q42. Which of the following is a sign of a positive team culture?

- A) Blaming individuals when a project fails
- B) Celebrating failures as learning opportunities and sharing credit for successes ✓**
- C) Competing internally to sabotage colleagues
- D) Withholding information to maintain personal advantage

■ **Positive team culture = psychological safety, shared accountability, and mutual support.**

Q43. 'Change management' in an IT project primarily addresses:

- A) Hardware upgrades only
- B) The people side of transitions — preparing and supporting individuals through change ✓**
- C) Version control in coding
- D) Financial budget changes
- **Change management focuses on human adoption of new processes, systems or structures.**

Q44. You notice a billing discrepancy in your project's financial report. You should:

- A) Adjust it yourself without telling anyone
- B) Bring it to your project manager's attention immediately ✓**
- C) Ignore small amounts
- D) Report it directly to the client without internal review
- **Financial discrepancies must be flagged internally immediately.**

Q45. Which statement best reflects TCS's commitment to sustainability?

- A) TCS focuses only on profit maximisation
- B) TCS integrates environmental and social responsibility into its business strategy ✓**
- C) Sustainability is only for manufacturing companies
- D) TCS does not publish ESG reports
- **TCS has strong ESG commitments including net-zero, education, and rural development.**

Q46. In a high-pressure project, a colleague suggests cutting corners on quality. You should:

- A) Agree to meet the deadline at any cost
- B) Raise the quality risk with your manager and explore alternatives like requesting timeline extension ✓**
- C) Stay silent
- D) Blame the client for unrealistic timelines
- **Quality shortcuts create long-term risks — escalate and explore options rather than compromising.**

Q47. 'Psychological safety' in a team means:

- A) Physical security of the office
- B) Feeling safe to speak up, share ideas, and admit mistakes without fear of punishment ✓**
- C) Only senior employees have the right to speak
- D) All decisions are made by the team lead
- **Psychological safety enables innovation, honest communication, and learning from mistakes.**

Q48. Which TCS policy governs the use of social media by employees?

- A) TCS Social Media Policy
- B) TCS IT Acceptable Use Policy
- C) Both — TCS has a Social Media Policy and AUP ✓**
- D) No policy exists
- **TCS has an IT Acceptable Use Policy and a Social Media Policy governing employee conduct online.**

Q49. What should you do before sharing any TCS or client data with a third-party vendor?

- A) Share freely as they are a vendor
- B) Verify the data classification, obtain necessary approvals, and ensure an NDA is in place
- C) Ask your colleague if it is okay ✓**
- D) Share only via WhatsApp to avoid email records
- **Data sharing with vendors requires classification check, approvals, and legal agreements.**

Q50. 'Proactive communication' means:

- A) Waiting for someone to ask before sharing updates
- B) Anticipating information needs and sharing relevant updates before being asked ✓**
- C) Sending excessive emails
- D) Only communicating in meetings
- **Proactive communication = volunteering timely, relevant information without waiting to be asked.**

■ SECTION C — UI: HTML, CSS & JavaScript Basics

50 Questions | Tags, Selectors, Box Model, Flexbox, JS Basics

Q1. Which HTML tag defines the document title shown in the browser tab?

- A) <head>
- B) <title> ✓**
- C) <meta>
- D) <header>

■ The <title> tag inside <head> sets the browser tab title.

Q2. What is the correct HTML5 doctype declaration?

- A) <!DOCTYPE HTML5>
- B) <!DOCTYPE html> ✓**
- C) <!DOCTYPE HTML PUBLIC>
- D) <html type=5>

■ HTML5 uses the simple <!DOCTYPE html> declaration.

Q3. Which attribute makes a text input mandatory in an HTML form?

- A) mandatory
- B) required ✓**
- C) validate
- D) must-fill

■ The 'required' attribute makes an input field mandatory for form submission.

Q4. Which CSS property controls the space between the element content and its border?

- A) margin
- B) gap
- C) padding ✓**
- D) spacing

■ Padding = space between content and border. Margin = space outside the border.

Q5. What does 'em' unit represent in CSS?

- A) Pixels
- B) Percentage of parent font-size ✓**
- C) A fixed unit equal to 1cm
- D) Viewport width

■ 1em = the font-size of the current/parent element.

Q6. Which CSS selector selects all <p> elements inside a <div>?

- A) div + p
- B) div p ✓**
- C) div > p
- D) div ~ p

■ 'div p' selects all <p> descendants (any depth) inside a div.

Q7. What is the default value of the CSS 'position' property?

- A) relative
- B) absolute
- C) fixed
- D) static ✓**

■ Default CSS position is 'static' — normal document flow.

Q8. Which HTML element is used for the largest heading?

- A) <h6>
- B) <h1>

C) <h1> ✓

D) <big>

■ <h1> is the largest heading; <h6> is the smallest.

Q9. In CSS Flexbox, which property defines the main axis direction?

A) align-items

B) justify-content

C) flex-direction ✓

D) flex-wrap

■ flex-direction sets main axis: row (default), column, row-reverse, column-reverse.

Q10. Which HTML tag creates an unordered (bulleted) list?

A)

B)

C) ✓

D) <list>

■ creates an unordered list; creates an ordered (numbered) list.

Q11. What does the CSS property 'z-index' control?

A) Element width

B) Zoom level

C) Stacking order of overlapping elements ✓

D) Font size

■ z-index controls which element appears on top when elements overlap.

Q12. Which input type validates an email format in HTML5?

A) text

B) email ✓

C) validate

D) contact

■ <input type='email'> automatically validates email format.

Q13. Which CSS property makes text bold?

A) text-style: bold

B) font-weight: bold ✓

C) font-style: bold

D) text-weight: bold

■ font-weight: bold (or 700) makes text bold.

Q14. What is the correct way to add an external CSS file to an HTML page?

A) <style src='style.css'>

B) <css href='style.css'>

C) <link rel='stylesheet' href='style.css'> ✓

D) <script src='style.css'>

■ <link rel='stylesheet' href='...'> links external CSS files.

Q15. Which HTML tag is used to embed JavaScript in a webpage?

A) <js>

B) <javascript>

C) <script> ✓

D) <code>

■ <script> tags are used to embed or reference JavaScript.

Q16. In CSS, 'display: none' differs from 'visibility: hidden' because:

A) They are identical

B) display:none removes the element from flow; visibility:hidden hides it but keeps the space ✓

C) visibility:hidden is faster

D) display:none only works on block elements

■ display:none removes from layout; visibility:hidden keeps space but hides visually.

Q17. What does the HTML 'alt' attribute on an tag provide?

- A) Image caption
- B) Alternative text if image fails to load (also for accessibility) ✓**
- C) Image title
- D) Image tooltip

■ Alt text is crucial for accessibility (screen readers) and when images fail to load.

Q18. Which CSS pseudo-class targets an element when hovered over?

- A) :focus
- B) :active
- C) :hover ✓**
- D) :visited

■ :hover applies styles when the user's pointer is over the element.

Q19. What is 'responsive design' in web development?

- A) A website that responds to user clicks quickly
- B) A design that adapts layout to different screen sizes using CSS ✓**
- C) A fast-loading website
- D) A design with animated responses

■ Responsive design uses flexible grids, images, and media queries to adapt to screen sizes.

Q20. Which CSS unit is relative to the viewport width?

- A) em
- B) px
- C) vw ✓**
- D) rem

■ vw = 1% of the viewport width. vh = 1% of viewport height.

Q21. What does the HTML <section> tag represent?

- A) A page header
- B) A standalone thematic section of content ✓**
- C) An external link
- D) A form section

■ <section> defines a thematic grouping of content, typically with a heading.

Q22. Which CSS property is used to add shadow to an element?

- A) text-shadow
- B) element-shadow
- C) box-shadow ✓**
- D) shadow

■ box-shadow adds shadow to the box of an element. text-shadow is for text only.

Q23. In JavaScript, which method adds an element to the end of an array?

- A) push() ✓**
- B) append()
- C) add()
- D) insert()

■ Array.push() adds one or more elements to the end of an array.

Q24. What does 'CSS specificity' determine?

- A) Loading speed of CSS
- B) Which CSS rule applies when multiple rules target the same element ✓**
- C) Order of HTML elements
- D) Browser compatibility

■ Specificity determines which conflicting CSS rule wins. ID > Class > Element.

Q25. Which HTML tag defines a table row?

- A) <td>

B) <th>

C) <tr> ✓

D) <table-row>

■ <tr> = table row. <td> = table data cell. <th> = table header cell.

Q26. Which CSS property controls text alignment?

A) align

B) text-align ✓

C) justify

D) font-align

■ **text-align: left/right/center/justify** controls horizontal text alignment.

Q27. What is the purpose of the HTML <meta charset='UTF-8'> tag?

A) Sets page language

B) Declares character encoding for the page ✓

C) Adds metadata for SEO

D) Sets page width

■ **charset='UTF-8'** ensures proper rendering of special characters across languages.

Q28. In CSS Grid, which property defines the number of columns?

A) grid-columns

B) grid-template-columns ✓

C) column-count

D) flex-columns

■ **grid-template-columns** defines the column structure in CSS Grid.

Q29. Which JavaScript event fires when a user clicks an element?

A) onhover

B) onchange

C) onclick ✓

D) onfocus

■ **onclick** fires when an element is clicked.

Q30. What does 'float: left' do in CSS?

A) Moves element to the bottom

B) Removes element from normal flow and aligns it to the left ✓

C) Makes text italic

D) Centres the element

■ **float** pulls an element out of normal flow and positions it left/right with text wrapping.

Q31. Which HTML5 tag is used for navigation links?

A) <links>

B) <nav> ✓

C) <menu>

D) <navigation>

■ **<nav>** is the semantic HTML5 element for navigation menus.

Q32. What is the CSS 'overflow: hidden' property used for?

A) Hides the entire element

B) Clips content that extends beyond the element's box ✓

C) Adds a scrollbar

D) Makes text invisible

■ **overflow:hidden** clips any content that overflows outside the element boundary.

Q33. In JavaScript, 'typeof null' returns:

A) 'null'

B) 'object' ✓

C) 'undefined'

D) 'empty'

■ **typeof null === 'object'** — a well-known JavaScript quirk/bug.

Q34. Which CSS property sets the minimum width of an element?

A) min-width ✓

B) width-min

C) base-width

D) flex-min

■ **min-width** ensures an element is never smaller than the specified value.

Q35. What does the 'rel' attribute in <link rel='stylesheet'> specify?

A) File path

B) Relationship between document and linked resource ✓

C) CSS version

D) Browser compatibility

■ **The 'rel' attribute defines the relationship — 'stylesheet' tells the browser it is CSS.**

Q36. Which JavaScript method converts a JSON string to an object?

A) JSON.parse() ✓

B) JSON.stringify()

C) JSON.convert()

D) JSON.decode()

■ **JSON.parse()** deserialises a JSON string into a JavaScript object.

Q37. Which CSS property adds rounded corners to an element?

A) corner-radius

B) border-round

C) border-radius ✓

D) round-corners

■ **border-radius** controls the rounding of an element's corners.

Q38. What is the 'DOM' in web development?

A) Document Object Model — a tree representation of the HTML page ✓

B) Data Object Mapping

C) Display Output Mode

D) Dynamic Output Method

■ **DOM = Document Object Model — JavaScript uses it to manipulate HTML/CSS dynamically.**

Q39. Which HTML attribute opens a link in a new tab?

A) target='_new'

B) target='_blank' ✓

C) open='tab'

D) href='newtab'

■ **target='_blank'** opens the hyperlink in a new browser tab.

Q40. What does 'CSS reset' accomplish?

A) Clears all JavaScript

B) Removes default browser styling for consistent cross-browser display ✓

C) Deletes all CSS

D) Resets the page language

■ **CSS reset overrides browser default styles to create a consistent baseline.**

Q41. In CSS, which selector has the highest specificity?

A) #id ✓

B) element

C) *

D) .class

■ **ID selectors (#id)** have higher specificity than class (.class) or element selectors.

Q42. Which HTML tag is used to define a cell in a table header row?

A) <td>

B) <tr>

C) <th> ✓

D) <thead>

■ **<th> = table header cell (bold and centred by default).**

Q43. What is 'lazy loading' in web performance?

- A) Loading all resources at page startup
 - B) Deferring loading of images/resources until they are needed ✓**
 - C) A slow website
 - D) Loading only CSS first
- **Lazy loading improves performance by deferring off-screen resources.**

Q44. Which CSS value for 'position' removes the element from normal flow and positions relative to nearest positioned ancestor?

- A) static
 - B) relative
 - C) absolute ✓**
 - D) fixed
- **position:absolute removes from normal flow and positions relative to the nearest non-static ancestor.**

Q45. In JavaScript, which operator checks both value AND type equality?

- A) ==
 - B) === ✓**
 - C) =
 - D) !=
- **=== is strict equality — checks both value and type. == only checks value.**

Q46. Which HTML5 input type shows a colour picker?

- A) type='colour'
 - B) type='color' ✓**
 - C) type='rgb'
 - D) type='picker'
- **<input type='color'> renders a native colour picker widget.**

Q47. Which CSS property controls the space between flex items?

- A) spacing
 - B) item-gap
 - C) gap ✓**
 - D) flex-gap
- **gap (formerly grid-gap) controls spacing between flex and grid items.**

Q48. What is 'semantic HTML'?

- A) HTML with lots of divs
 - B) Using HTML elements that convey meaning about the content (e.g. <article>, <nav>) ✓**
 - C) HTML without CSS
 - D) HTML with inline styles only
- **Semantic HTML uses meaningful elements that describe their purpose, improving accessibility and SEO.**

Q49. Which JavaScript method removes and returns the last element of an array?

- A) shift()
 - B) splice()
 - C) pop() ✓**
 - D) remove()
- **Array.pop() removes and returns the last element.**

Q50. What does 'viewport meta tag' do?

- A) Sets server location
 - B) Controls how page scales on mobile devices ✓**
 - C) Defines page author
 - D) Sets browser language
- **<meta name='viewport' content='width=device-width, initial-scale=1'> controls mobile scaling.**

■ SECTION D — Unix / Linux Commands

50 Questions | Commands, Permissions, Shell, File System

Q1. Which command displays the current working directory?

- A) cwd
- B) dir
- C) pwd ✓**
- D) path

■ **pwd = Print Working Directory.**

Q2. Which command lists files including hidden files?

- A) ls
- B) ls -l
- C) ls -a ✓**
- D) ls -h

■ **ls -a shows all files including hidden (starting with dot).**

Q3. What does 'chmod 755 file.sh' set?

- A) rwxrwxrwx
- B) rwxr-xr-x ✓**
- C) rwxrw-rw-
- D) rwx-----

■ **7=rwx(owner), 5=r-x(group), 5=r-x(others) → rwxr-xr-x.**

Q4. Which command searches for a pattern in a file?

- A) find
- B) locate
- C) grep ✓**
- D) awk

■ **grep searches for text patterns within files.**

Q5. What does the pipe '|' operator do?

- A) Redirects to file
- B) Passes output of one command as input to another ✓**
- C) Appends to file
- D) Runs commands in background

■ **Pipe chains commands: output of left → input of right.**

Q6. Which command shows running processes?

- A) list
- B) jobs
- C) ps ✓**
- D) proc

■ **ps (process status) lists running processes. ps aux for all.**

Q7. What does 'rm -rf dir' do?

- A) Renames directory
- B) Removes directory and all contents recursively and forcefully ✓**
- C) Creates directory
- D) Moves directory

■ **rm -rf forcefully and recursively deletes a directory and its contents.**

Q8. Which command moves or renames a file?

- A) copy
- B) cp

C) mv ✓

D) rename

■ mv = move or rename files/directories.

Q9. What does 'cat file.txt' do?

A) Creates file

B) Deletes file

C) Displays file contents ✓

D) Edits file

■ cat concatenates and displays file contents to standard output.

Q10. Which symbol appends output to a file without overwriting?

A) <

B) >> ✓

C) >

D) &

■ >> appends output to file. > overwrites the file.

Q11. Which command creates a new directory?

A) touch

B) mkdir ✓

C) newdir

D) create

■ mkdir = make directory.

Q12. What does 'wc -l file.txt' count?

A) Words

B) Characters

C) Lines ✓

D) Bytes

■ wc -l counts lines. wc -w counts words. wc -c counts bytes.

Q13. Which command displays the last 10 lines of a file?

A) head

B) last

C) tail ✓

D) end

■ tail -n 10 (default) shows last 10 lines. head shows first 10.

Q14. What does 'chmod +x file.sh' do?

A) Removes execute permission

B) Adds execute permission to all users ✓

C) Changes file owner

D) Makes file read-only

■ chmod +x adds execute permission for all (owner, group, others).

Q15. Which command finds files by name?

A) locate -name

B) grep -f

C) find . -name ✓

D) search -f

■ find . -name 'filename' searches recursively from current directory.

Q16. What does 'echo \$HOME' print?

A) Literal text \$HOME

B) Current user's home directory path ✓

C) Current directory

D) System hostname

■ \$HOME is an environment variable holding the home directory path.

Q17. Which command shows disk usage of a directory?

- A) df
- B) du ✓**
- C) disk
- D) ls -s

■ **du (disk usage) shows size of directories. df shows filesystem usage.**

Q18. What does 'chown user file' do?

- A) Changes file permissions
- B) Changes file owner to 'user' ✓**
- C) Copies file
- D) Renames file

■ **chown = change owner. chown user:group file to change both.**

Q19. Which command copies a file?

- A) mv
- B) copy
- C) cp ✓**
- D) ln

■ **cp = copy. cp -r for recursive directory copy.**

Q20. What does 'ps aux | grep apache' do?

- A) Starts Apache server
- B) Lists all processes and filters for 'apache' ✓**
- C) Kills Apache
- D) Installs Apache

■ **Combines ps aux (all processes) with grep to filter for 'apache'.**

Q21. Which command shows the first 5 lines of a file?

- A) tail -5
- B) head -5 ✓**
- C) cat -5
- D) top -5

■ **head -5 shows first 5 lines. head -n 5 also works.**

Q22. What is a 'symlink'?

- A) A compressed file
- B) A symbolic link — a pointer/shortcut to another file ✓**
- C) A system file
- D) A log file

■ **ln -s target link_name creates a symbolic link.**

Q23. Which command shows all environment variables?

- A) env ✓**
- B) vars
- C) printenv all
- D) set -e

■ **env lists all current environment variables.**

Q24. What does 'kill -9 PID' do?

- A) Pauses a process
- B) Terminates a process forcefully (SIGKILL) ✓**
- C) Restarts a process
- D) Lists processes

■ **kill -9 sends SIGKILL — forceful termination that cannot be caught.**

Q25. Which permission number gives read and write but NOT execute?

- A) 7

B) 6 ✓

C) 5

D) 4

■ $rw- = 4+2 = 6$. $rwX=7$, $r-x=5$, $r--=4$.

Q26. What does 'sudo' stand for?

- A) Super User Download
- B) Switch User Do
- C) Super User Do ✓**
- D) System Update Do

■ **sudo = Super User Do** — executes a command with elevated (root) privileges.

Q27. Which command shows network interfaces and IP addresses?

- A) ipconfig
- B) networkstat
- C) ifconfig / ip addr ✓**
- D) netif

■ **ifconfig (older) or ip addr (modern Linux)** displays network interface info.

Q28. What does 'history' command show?

- A) Browser history
- B) Previously executed shell commands ✓**
- C) File modification history
- D) System log history

■ **history** shows the list of previously entered shell commands.

Q29. Which command opens a manual page for a command?

- A) help
- B) info
- C) man ✓**
- D) docs

■ **man** command shows the manual (documentation) page for any command.

Q30. What does 'df -h' show?

- A) Directory files in human-readable format
- B) Disk filesystem usage in human-readable format ✓**
- C) Hidden files
- D) Default files

■ **df -h** shows disk space usage with human-readable sizes (KB/MB/GB).

Q31. Which command monitors real-time system resource usage?

- A) ps
- B) htop / top ✓**
- C) usage
- D) sysmon

■ **top** or **htop** shows live CPU, memory and process usage.

Q32. What does 'tar -czf archive.tar.gz dir/' do?

- A) Extracts a tar file
- B) Creates a gzip-compressed tar archive of dir/ ✓**
- C) Deletes dir/
- D) Lists contents of archive

■ **tar -czf = create(-c), gzip(-z), file(-f).** Used to compress directories.

Q33. Which command extracts a tar.gz archive?

- A) tar -czf
- B) tar -xzf ✓**
- C) tar -tzf
- D) unzip

■ **tar -xzf = extract(-x), gzip(-z), file(-f).**

Q34. What does 'crontab -e' do?

- A) Shows cron logs
- B) Edits the cron job schedule for the current user ✓**

C) Runs all cron jobs immediately

D) Disables cron

■ **crontab -e opens the cron table editor to schedule recurring tasks.**

Q35. Which file stores user account information in Linux?

A) /etc/group

B) /etc/passwd ✓

C) /etc/users

D) /var/users

■ **/etc/passwd stores basic user account info (username, UID, home dir, shell).**

Q36. What does 'sort -r file.txt' do?

A) Sorts file alphabetically

B) Sorts file in reverse order ✓

C) Removes duplicates

D) Randomises order

■ **sort -r reverses the sort order. Default sort is ascending.**

Q37. Which command shows unique lines in a sorted file?

A) unique

B) uniq ✓

C) distinct

D) nodup

■ **uniq filters adjacent duplicate lines. Use sort file | uniq for full de-duplication.**

Q38. What does 'which python3' show?

A) Python version

B) Full path to the python3 executable ✓

C) Python documentation

D) Python packages installed

■ **which command shows the full path of the executable for a command.**

Q39. What is the purpose of '#!/bin/bash' at the top of a shell script?

A) A comment

B) Tells the OS to use bash to execute the script (shebang) ✓

C) Imports bash library

D) Sets script permissions

■ **Shebang line (!) tells the kernel which interpreter to use.**

Q40. Which command shows the last login time of users?

A) lastlog ✓

B) whoami

C) users

D) w

■ **lastlog shows the most recent login for all users.**

Q41. What does 'ln -s /etc/hosts myhosts' create?

A) A hard link

B) A copy of /etc/hosts

C) A symbolic link named 'myhosts' pointing to /etc/hosts ✓

D) A directory

■ **ln -s creates a symbolic (soft) link.**

Q42. Which command changes the current user's password?

A) passwd ✓

B) chpasswd

C) usermod

D) sudo password

■ **passwd changes the password for the current (or specified) user.**

Q43. What does 'nohup command &' do?

- A) Runs command in foreground
 - B) Runs command in background, immune to hangup signals ✓**
 - C) Cancels a command
 - D) Mutes command output
- nohup + & runs a command in background that persists after logout.

Q44. Which command displays the hostname of the system?

- A) sysname
 - B) hostname ✓**
 - C) uname -n
 - D) host
- hostname displays or sets the system's network name.

Q45. What does 'awk '{print \$2}' file.txt' do?

- A) Prints first column
 - B) Prints second column/field of each line ✓**
 - C) Prints last column
 - D) Deletes second column
- awk uses \$1, \$2... to access whitespace-separated fields. \$2 = second field.

Q46. Which command checks if a remote host is reachable?

- A) tracet
 - B) netstat
 - C) ping ✓**
 - D) curl
- ping sends ICMP packets to test network connectivity to a host.

Q47. What is '/dev/null' in Linux?

- A) A device file
 - B) A null device that discards all data written to it ✓**
 - C) A directory for null files
 - D) A RAM disk
- /dev/null is a special file that discards all input — used to suppress output.

Q48. Which command shows the size of a specific file in human-readable format?

- A) ls -h filename
 - B) du -sh filename ✓**
 - C) size filename
 - D) stat -h filename
- du -sh filename shows disk usage in human-readable format.

Q49. What does 'sed 's/old/new/g' file.txt' do?

- A) Deletes 'old' from file
 - B) Replaces first occurrence of 'old' with 'new'
 - C) Replaces all occurrences of 'old' with 'new' in file ✓**
 - D) Searches for 'old' and prints matching lines
- sed 's/old/new/g' — the 'g' flag replaces ALL occurrences globally.

Q50. Which command shows all currently logged-in users?

- A) users
 - B) who
 - C) w
 - D) all of the above ✓**
- users, who, and w all show logged-in users (with varying levels of detail).

■ ■ SECTION E — Java & Python MCQs

50 Questions | OOP, Data Structures, Logic, Syntax

Q1. What is the output of: `print(type({}))`?

A) `<class 'list'>`

B) `<class 'dict'>` ✓

C) `<class 'set'>`

D) `<class 'tuple'>`

■ `{}` creates an empty dict. `type({}) = <class 'dict'>`.

Q2. Which Python keyword defines a generator function?

A) `return`

B) `generate`

C) `yield` ✓

D) `async`

■ `yield` turns a function into a generator, enabling lazy evaluation.

Q3. What is the output of: `print(3 * '2')`?

A) 6

B) 222 ✓

C) Error

D) '222'

■ `String * int` in Python repeats the string. `'2' * 3 = '222'`.

Q4. Which Python method removes whitespace from both ends of a string?

A) `strip()` ✓

B) `trim()`

C) `remove()`

D) `clean()`

■ `str.strip()` removes leading and trailing whitespace.

Q5. What is a Python 'lambda'?

A) A loop construct

B) An anonymous inline function ✓

C) A class decorator

D) A module import

■ `lambda` args: expression — a one-line anonymous function.

Q6. What does '`list(range(1, 6))`' produce?

A) [1,2,3,4,5,6]

B) [1,2,3,4,5] ✓

C) [0,1,2,3,4,5]

D) [2,3,4,5,6]

■ `range(1,6) = 1 to 5 inclusive. End is exclusive.`

Q7. Which of the following creates a set in Python?

A) `s = []`

B) `s = {}`

C) `s = {1,2,3}` ✓

D) `s = (1,2,3)`

■ A set uses curly braces with elements: {1,2,3}. {} alone creates a dict.

Q8. What is the output of: `x = [1,2,3]; print(x[-1])`?

A) 1

B) 3 ✓

C) Error

D) -1

■ **Negative indexing in Python: `x[-1]` = last element = 3.**

Q9. Which Python exception is raised for division by zero?

A) ArithmeticError

B) ZeroDivisionError ✓

C) MathError

D) DivisionError

■ **ZeroDivisionError is raised when dividing by zero.**

Q10. What does 'dict.get(key, default)' do?

A) Raises error if key missing

B) Returns value for key, or default if key not found ✓

C) Deletes key

D) Updates key

■ **dict.get() safely retrieves a value without raising KeyError.**

Q11. What is the output of: `print(bool(0), bool(""), bool([]))`?

A) True True True

B) False False False ✓

C) False True False

D) True False True

■ **0, empty string, and empty list are all falsy in Python.**

Q12. Which Python decorator makes a method belong to the class, not an instance?

A) @property

B) @instancemethod

C) @classmethod / @staticmethod ✓

D) @abstract

■ **@staticmethod and @classmethod both define class-level methods.**

Q13. What is the correct way to open a file for writing in Python?

A) `open('f.txt','r')`

B) `open('f.txt','w')` ✓

C) `open('f.txt','x')`

D) `open('f.txt','a')`

■ **'w' mode opens for writing (creates or overwrites). 'a' appends.**

Q14. What does 'enumerate([10,20,30])' return when iterated?

A) [0,10,1,20,2,30]

B) (0,10),(1,20),(2,30) pairs ✓

C) [10,20,30] with indices

D) Error

■ **enumerate returns (index, value) pairs.**

Q15. Which of the following is NOT a valid Python data type?

A) list

B) tuple

C) array ✓

D) dict

■ **Python does not have a built-in 'array' type (lists serve this purpose; array module exists but is separate).**

Q16. What is the output of: `print(10 // 3)`?

A) 3.33

B) 3 ✓

C) 4

D) Error

■ **// is floor division. `10//3 = 3` (truncates decimal).**

Q17. Which keyword is used to inherit a class in Python?

- A) extends
- B) implements
- C) class Child(Parent): ✓**
- D) inherits

■ Python uses class Child(Parent): syntax for inheritance.

Q18. What does 'map(func, iterable)' do?

- A) Filters elements
- B) Applies func to each element and returns an iterator ✓**
- C) Sorts elements
- D) Reduces to single value

■ map() applies a function to every item in an iterable.

Q19. What is the output of: 'hello'.upper()[:3]?

- A) hel
- B) HEL ✓**
- C) HEL
- D) Hel

■ 'hello'.upper() = 'HELLO'. [:3] = 'HEL'.

Q20. Which Python module is used for regular expressions?

- A) regex
- B) re ✓**
- C) regexp
- D) pattern

■ The 're' module provides regular expression operations in Python.

Q21. What does 'zip([1,2],[3,4])' produce when converted to list?

- A) [1,2,3,4]
- B) [(1,3),(2,4)] ✓**
- C) [[1,3],[2,4]]
- D) Error

■ zip pairs elements: [(1,3),(2,4)].

Q22. What is a Python 'list comprehension'?

- A) A way to compress lists
- B) A concise syntax to create lists: [expr for item in iterable] ✓**
- C) A method to sort lists
- D) A list of methods

■ List comprehension: [x*2 for x in range(5)] = [0,2,4,6,8].

Q23. Which built-in function returns the length of an object?

- A) size()
- B) count()
- C) len() ✓**
- D) length()

■ len() returns the number of items in a sequence or collection.

Q24. What does 'try...except...finally' ensure?

- A) Catches only runtime errors
- B) The finally block always executes regardless of exception ✓**
- C) Skips errors silently
- D) Restarts the program

■ finally always runs — used for cleanup like closing files.

Q25. What is the output of: print([x2 for x in range(4)])?**

- A) [1,4,9,16]

B) [0,1,4,9] ✓

C) [0,1,2,3]

D) [1,2,3,4]

■ `range(4)` = 0,1,2,3. Squared: 0,1,4,9.

Q26. In Java, which keyword prevents a variable from being reassigned?

- A) static
- B) private
- C) final ✓**
- D) const

■ final variables cannot be reassigned after initialisation.

Q27. What is the output of: System.out.println(5 + 3 + 'a')?

- A) Error
- B) 8a
- C) 105 ✓**
- D) '8a'

■ 'a' ASCII = 97. 5+3=8, 8+97=105. Java char is treated as int in arithmetic.

Q28. Which Java collection maintains insertion order and allows duplicates?

- A) HashSet
- B) TreeSet
- C) ArrayList ✓**
- D) LinkedHashMap

■ ArrayList maintains insertion order and allows duplicate elements.

Q29. What does 'public static void main(String[] args)' mean?

- A) A method visible only in the class
- B) The program entry point — public, class-level, no return, takes string array ✓**
- C) A constructor
- D) A static variable

■ main() is the entry point of every Java application.

Q30. Which principle of OOP allows a subclass to provide a specific implementation of a method already defined in its superclass?

- A) Encapsulation
- B) Abstraction
- C) Method Overriding ✓**
- D) Method Overloading

■ Method Overriding = subclass redefines a superclass method with @Override.

Q31. What is the difference between '==' and '.equals()' in Java for Strings?

- A) They are identical
- B) == compares references; .equals() compares content ✓**
- C) == is faster
- D) equals() only works for primitives

■ '==' checks if two references point to the same object. .equals() compares string content.

Q32. Which Java keyword is used to call the parent class constructor?

- A) parent()
- B) super() ✓**
- C) base()
- D) this()

■ super() calls the parent class constructor from a subclass.

Q33. Which exception is thrown when an array is accessed out of bounds?

- A) IndexOutOfBoundsException
- B) ArrayException
- C) OutOfRangeException
- D) ArrayIndexOutOfBoundsException ✓**

■ ArrayIndexOutOfBoundsException is thrown for invalid array index access.

Q34. What is an 'interface' in Java?

- A) A class with only static methods

B) A blueprint with abstract methods that a class must implement ✓

C) A type of exception

D) A database connector

■ **Interface = contract of method signatures that implementing classes must fulfil.**

Q35. Which Java class is used for mutable strings?

A) String

B) StringBuffer / StringBuilder ✓

C) MutableString

D) StringArray

■ **StringBuilder (non-threadsafe) and StringBuffer (threadsafe) allow string mutation.**

Q36. What does 'ArrayList vs LinkedList' differ in?

A) Size limit

B) ArrayList uses dynamic array (fast random access); LinkedList uses nodes (fast insert/delete) ✓

C) Sorting algorithm

D) Data types supported

■ **ArrayList is better for random access; LinkedList is better for frequent insertions/deletions.**

Q37. What is the output of: int x = 5; System.out.println(x++)?

A) 6

B) 5 ✓

C) Error

D) 4

■ **Post-increment: x++ returns current value (5), then increments x to 6.**

Q38. Which keyword prevents a class from being subclassed in Java?

A) private

B) static

C) final ✓

D) abstract

■ **final class cannot be extended (e.g., java.lang.String is final).**

Q39. What is 'polymorphism' in OOP?

A) One class inheriting another

B) The ability of one interface to be used for different underlying data types ✓

C) Hiding implementation details

D) Breaking a class into smaller units

■ **Polymorphism = 'many forms' — one interface, multiple implementations.**

Q40. Which Java data structure is LIFO (Last In First Out)?

A) Queue

B) Stack ✓

C) ArrayList

D) HashMap

■ **Stack is LIFO. Queue is FIFO.**

Q41. What is 'garbage collection' in Java?

A) Deleting unused imports

B) Automatic memory management — JVM reclaims unreferenced heap objects ✓

C) Manual memory deallocation

D) Clearing console output

■ **Java's GC automatically frees memory occupied by unreachable objects.**

Q42. Which Java keyword is used for exception handling?

A) catch ✓

B) handle

C) rescue

D) trap

■ Java uses try-catch-finally for exception handling.

Q43. What is the access level of 'protected' in Java?

- A) Same class only
 - B) Same class and subclasses (and same package) ✓**
 - C) All classes
 - D) Only subclasses in different packages
- **protected = same package + subclasses (even in different packages).**

Q44. What does 'HashMap.put(key, value)' do?

- A) Retrieves a value
 - B) Inserts or updates a key-value pair ✓**
 - C) Removes a key
 - D) Checks if key exists
- **put() inserts a new key-value pair or updates the value if key already exists.**

Q45. Which annotation suppresses compiler warnings about type safety in Java?

- A) @Override
 - B) @SuppressWarnings ✓**
 - C) @FunctionalInterface
 - D) @Deprecated
- **@SuppressWarnings('unchecked') suppresses type-safety warnings.**

Q46. What is a 'constructor' in Java?

- A) A method that returns a value
 - B) A special method called when an object is created, used for initialisation ✓**
 - C) A static method
 - D) An abstract method
- **Constructor = same name as class, no return type, called via 'new'.**

Q47. What is the time complexity of binary search?

- A) $O(n)$
 - B) $O(n^2)$
 - C) $O(\log n)$ ✓**
 - D) $O(1)$
- **Binary search halves the search space each step → $O(\log n)$.**

Q48. Which Java collection class does NOT allow null keys?

- A) HashMap
 - B) LinkedHashMap
 - C) Hashtable ✓**
 - D) TreeMap (null keys disallowed)
- **Hashtable does not allow null keys or null values (unlike HashMap).**

Q49. What is 'method overloading' in Java?

- A) A method calling itself
 - B) Multiple methods with same name but different parameter lists in the same class ✓**
 - C) A method in a subclass overriding parent
 - D) Using the same method in two classes
- **Overloading = same method name, different signatures in one class.**

Q50. What is the output of: System.out.println(1 + 2 + '3')?

- A) 123
 - B) 6 ✓**
 - C) Error
 - D) 33
- **1+2=3, '3' ASCII=51. 3+51=54... Wait: actually '3'=51, so 1+2=3, 3+51=54. Answer is 54. However in common TCS pattern: correct answer is 54.**

■ ■ SECTION F — SQL / MySQL / PL-SQL

50 Questions | Queries, Joins, Functions, Stored Procedures

Q1. Which SQL clause filters rows BEFORE grouping?

- A) HAVING
- B) ORDER BY
- C) WHERE ✓**
- D) FILTER

■ **WHERE filters individual rows before GROUP BY. HAVING filters groups after.**

Q2. What does 'SELECT DISTINCT city FROM employees' do?

- A) Returns all cities including duplicates
- B) Returns unique cities only ✓**
- C) Returns city counts
- D) Deletes duplicate cities

■ **DISTINCT eliminates duplicate values from the result set.**

Q3. Which JOIN returns only matching rows from both tables?

- A) LEFT JOIN
- B) RIGHT JOIN
- C) INNER JOIN ✓**
- D) FULL OUTER JOIN

■ **INNER JOIN returns only rows where the join condition matches in BOTH tables.**

Q4. What does the 'COUNT(*)' function return?

- A) Sum of values
- B) Number of rows ✓**
- C) Maximum value
- D) Average value

■ **COUNT(*) counts all rows including NULLs.**

Q5. Which SQL command is used to permanently delete a table and its structure?

- A) DELETE
- B) TRUNCATE
- C) DROP ✓**
- D) REMOVE

■ **DROP TABLE permanently removes the table and its structure. DELETE removes rows only.**

Q6. What is the difference between DELETE and TRUNCATE?

- A) They are identical
- B) DELETE removes specific rows (with WHERE); TRUNCATE removes all rows and is faster ✓**
- C) TRUNCATE supports WHERE clause
- D) DELETE removes the table structure

■ **DELETE is DML (can rollback, supports WHERE). TRUNCATE is DDL (faster, no WHERE).**

Q7. Which clause is used to sort results?

- A) GROUP BY
- B) SORT BY
- C) ORDER BY ✓**
- D) ARRANGE BY

■ **ORDER BY column_name [ASC|DESC] sorts query results.**

Q8. What does a PRIMARY KEY constraint ensure?

- A) Allows NULL values
- B) Each value is unique and NOT NULL ✓**

C) Values can be duplicated

D) Allows multiple NULLs

■ **PRIMARY KEY = UNIQUE + NOT NULL. Each table can have only one primary key.**

Q9. Which function returns the current date in SQL?

A) NOW()

B) GETDATE() / CURDATE() ✓

C) DATE()

D) CURRENT()

■ **MySQL: CURDATE() or NOW() (with time). SQL Server: GETDATE().**

Q10. What is a FOREIGN KEY?

A) A key that is always unique

B) A column that references a PRIMARY KEY in another table ✓

C) An index key

D) A composite key

■ **FOREIGN KEY enforces referential integrity by linking to another table's PK.**

Q11. Which SQL aggregate function returns the total sum?

A) COUNT

B) AVG

C) SUM ✓

D) MAX

■ **SUM() adds up all values in a numeric column.**

Q12. What does 'GROUP BY department' do?

A) Filters by department

B) Sorts by department

C) Groups rows with same department value for aggregation ✓

D) Creates a new column

■ **GROUP BY groups rows with the same value, used with aggregate functions.**

Q13. Which of the following is a DDL (Data Definition Language) command?

A) SELECT

B) INSERT

C) CREATE ✓

D) UPDATE

■ **DDL commands: CREATE, ALTER, DROP, TRUNCATE. DML: SELECT, INSERT, UPDATE, DELETE.**

Q14. What does 'HAVING COUNT(*) > 5' do in a query?

A) Filters individual rows with more than 5 columns

B) Filters groups having more than 5 rows ✓

C) Returns only 5 rows

D) Counts rows named HAVING

■ **HAVING filters aggregated groups — cannot use WHERE for aggregate conditions.**

Q15. Which SQL keyword adds new rows to a table?

A) ADD

B) INSERT INTO ✓

C) UPDATE

D) APPEND

■ **INSERT INTO table (cols) VALUES (vals) adds new rows.**

Q16. What does NULL represent in SQL?

A) Zero

B) Empty string

C) Unknown or missing value ✓

D) False

■ **NULL means the absence of a value — not 0 or empty string.**

Q17. Which operator checks for NULL values?

- A) = NULL
- B) == NULL
- C) IS NULL ✓**
- D) NULL IS

■ Always use IS NULL or IS NOT NULL — never '= NULL'.

Q18. What is a 'subquery'?

- A) A stored procedure
- B) A query nested inside another query ✓**
- C) A view
- D) An index

■ Subquery = a query within another query, in WHERE, FROM, or SELECT clauses.

Q19. Which JOIN returns all rows from the LEFT table plus matching rows from the right?

- A) INNER JOIN
- B) FULL OUTER JOIN
- C) RIGHT JOIN
- D) LEFT JOIN ✓**

■ LEFT JOIN keeps ALL rows from left table; unmatched right rows show NULL.

Q20. What does 'UPDATE employees SET salary = salary * 1.1 WHERE dept = 'IT' do?

- A) Deletes IT employees
- B) Gives IT employees a 10% raise ✓**
- C) Sets IT salary to 1.1
- D) Adds a new IT employee

■ salary * 1.1 multiplies current salary by 1.1 = 10% increase.

Q21. Which SQL statement creates a virtual table based on a query?

- A) TABLE
- B) INDEX
- C) VIEW ✓**
- D) PROCEDURE

■ CREATE VIEW name AS SELECT... creates a virtual table/view.

Q22. What is a 'stored procedure' in SQL?

- A) A saved SELECT query
- B) A precompiled set of SQL statements stored and executed on the server ✓**
- C) An index
- D) A trigger

■ Stored procedures are precompiled SQL blocks that can accept parameters and be reused.

Q23. Which clause limits the number of rows returned?

- A) TOP / LIMIT ✓**
- B) FIRST
- C) ROWNUM only
- D) RESTRICT

■ LIMIT (MySQL/PostgreSQL) or TOP (SQL Server) limits rows returned.

Q24. What does 'ALTER TABLE employees ADD COLUMN age INT' do?

- A) Deletes the age column
- B) Creates a new table
- C) Adds an 'age' column to the employees table ✓**
- D) Renames the table

■ ALTER TABLE ADD COLUMN adds a new column to an existing table.

Q25. Which SQL function returns the length of a string?

- A) SIZE()

B) LEN() / LENGTH() ✓

C) COUNT()

D) STR_LEN()

■ LEN() in SQL Server; LENGTH() in MySQL — both return string character count.

Q26. What is a 'trigger' in SQL?

A) A scheduled job

B) Automatic SQL code that executes before/after INSERT, UPDATE, or DELETE ✓

C) An index type

D) A foreign key

■ Triggers auto-execute on DML events (INSERT/UPDATE/DELETE).

Q27. Which of the following is a scalar function in SQL?

A) GROUP BY

B) SUM()

C) UPPER() ✓

D) COUNT(*)

■ UPPER() is a scalar function — operates on a single value per row.

Q28. What does 'COALESCE(col1, col2, 'default')' return?

A) First non-NULL value from the list ✓

B) NULL always

C) Sum of col1 and col2

D) An error

■ COALESCE returns the first non-NULL argument in the list.

Q29. Which command grants SELECT permission to a user in SQL?

A) ALLOW SELECT ON table TO user

B) GRANT SELECT ON table TO user ✓

C) PERMIT SELECT

D) ADD PERMISSION SELECT

■ GRANT privilege ON object TO user — standard SQL privilege management.

Q30. What does 'ROLLBACK' do in SQL?

A) Saves all transactions

B) Undoes all changes since the last COMMIT ✓

C) Commits and then clears

D) Deletes the database

■ ROLLBACK undoes uncommitted transactions in the current transaction block.

Q31. Which SQL clause is used for pattern matching?

A) MATCH

B) FIND

C) LIKE ✓

D) REGEX

■ LIKE uses wildcards: % (any characters) and _ (single character).

Q32. What does 'SELECT MAX(salary) FROM employees WHERE dept='Sales' return?

A) Average salary in Sales

B) Highest salary in Sales department ✓

C) All salaries in Sales

D) Count of Sales employees

■ MAX() returns the highest value — filtered by WHERE before aggregation.

Q33. What is 'normalisation' in database design?

A) Increasing data redundancy

B) Organising data to reduce redundancy and improve data integrity ✓

C) Encrypting a database

D) Adding more indexes

■ Normalisation (1NF, 2NF, 3NF) eliminates redundancy and ensures data integrity.

Q34. Which join returns all rows from BOTH tables including non-matching rows?

A) INNER JOIN

B) LEFT JOIN

C) RIGHT JOIN

D) FULL OUTER JOIN ✓

■ **FULL OUTER JOIN = LEFT JOIN + RIGHT JOIN** — all rows from both tables.

Q35. What does 'IN (val1, val2, val3)' do in a WHERE clause?

A) Checks if column value is within a range

B) Checks if column value matches any value in the list ✓

C) Inserts values

D) Checks inequality

■ **IN checks if the value matches ANY value in the specified list.**

Q36. Which constraint ensures a column value is unique but allows NULLs?

A) PRIMARY KEY

B) NOT NULL

C) UNIQUE ✓

D) CHECK

■ **UNIQUE allows one NULL (in most databases) while still enforcing uniqueness.**

Q37. What is a 'composite key'?

A) A key with encryption

B) A primary key made up of two or more columns ✓

C) A foreign key with two references

D) A key that allows NULLs

■ **Composite key = primary key formed by combining two or more columns.**

Q38. Which SQL function rounds a number to the nearest integer?

A) CEIL()

B) FLOOR()

C) ROUND() ✓

D) TRUNC()

■ **ROUND(value, decimal_places) rounds to nearest. CEIL rounds up, FLOOR rounds down.**

Q39. What does 'BETWEEN 50 AND 100' do in a WHERE clause?

A) Excludes values 50 and 100

B) Returns values from 50 to 100 inclusive ✓

C) Returns values between but not including 50 and 100

D) Inserts values 50 to 100

■ **BETWEEN is inclusive on both ends: WHERE salary BETWEEN 50 AND 100.**

Q40. Which command saves the current transaction permanently?

A) SAVE

B) COMMIT ✓

C) PUSH

D) END TRANSACTION

■ **COMMIT permanently saves all changes made in the current transaction.**

Q41. What does 'INDEX' do in SQL?

A) Adds a primary key

B) Creates a data structure to speed up queries on a column ✓

C) Removes duplicate rows

D) Enforces foreign key

■ **Indexes improve query performance by allowing faster data retrieval.**

Q42. Which aggregate function calculates the average?

A) MEAN()

B) SUM()/COUNT()

C) AVG() ✓

D) MEDIAN()

■ **AVG() calculates the arithmetic mean of a numeric column.**

Q43. What is a 'cursor' in PL-SQL?

- A) A type of join
 - B) A pointer used to iterate through query result sets row by row ✓**
 - C) An index
 - D) A trigger
- Cursors allow row-by-row processing of a SELECT result set in PL/SQL.

Q44. Which SQL clause is used with GROUP BY to filter groups?

- A) WHERE
 - B) FILTER
 - C) HAVING ✓**
 - D) RESTRICT
- HAVING filters after GROUP BY. WHERE filters before grouping.

Q45. What does '%' wildcard match in a LIKE clause?

- A) Any single character
 - B) Any sequence of characters (including none) ✓**
 - C) Only alphabets
 - D) Numbers only
- % matches any sequence of zero or more characters in LIKE.

Q46. Which keyword prevents a column from having NULL values?

- A) UNIQUE
 - B) NOT NULL ✓**
 - C) DEFAULT
 - D) CHECK
- NOT NULL constraint forces a column to always have a value.

Q47. What does 'CROSS JOIN' produce?

- A) Only matching rows
 - B) The Cartesian product of two tables (all combinations) ✓**
 - C) NULL rows
 - D) Only non-matching rows
- CROSS JOIN returns every row from table1 combined with every row from table2.

Q48. Which function converts a string to uppercase in SQL?

- A) TOUPPER()
 - B) UPPER() ✓**
 - C) CAPITALIZE()
 - D) STR_UPPER()
- UPPER(column) converts string values to uppercase.

Q49. What is 'referential integrity'?

- A) Data is stored efficiently
 - B) Foreign key values must match existing primary key values (or be NULL) ✓**
 - C) All columns have indexes
 - D) Data is encrypted
- Referential integrity ensures FK values reference valid PK values in the parent table.

Q50. Which SQL statement is used to retrieve data?

- A) INSERT
 - B) UPDATE
 - C) SELECT ✓**
 - D) CREATE
- SELECT is the DQL (Data Query Language) command for retrieving data.

■ Exam Day Quick Tips:

- KYT: Read TCS Xplore modules completely — questions come directly from there.
- BizSkill: Always choose the option reflecting TCS values (Ethics, Integrity, Communication).
- UI: Focus on semantic HTML5 tags, CSS specificity rules, and Flexbox/Grid.
- Unix: Memorise 30 core commands — especially `chmod`, `grep`, `find`, `awk`, `sed`.
- Java/Python: OOP concepts (inheritance, polymorphism) and String/List operations are most tested.
- SQL: JOINS, GROUP BY + HAVING, and aggregate functions appear in every exam.
- No negative marking — attempt ALL questions. Time per MCQ: ~1 minute.